

Sequences and Series (Part II) Homework

① If the 5th term of an arithmetic sequence is -6, and the 8th term is 18, what is the 12th term?

② If the 23rd term of an arithmetic sequence is 8.5, and the 32nd term is -36.5, what is the 1st term?

③ If the 10th term of a geometric sequence is -2, and the 15th term is -486, what is the 19th term?

④ If the 3rd term of a geometric sequence is $\frac{3}{4}$, and the 7th term is $\frac{1}{108}$, what is the 12th term? Assume that the common ratio is positive.

Find the sum of all of the terms in each geometric sequence.

⑤ $-\frac{3}{4}, -\frac{1}{2}, -\frac{1}{3}, \dots$ ⑥ $2, 1.4, 0.98, \dots$

⑦ $15, -6, \frac{12}{5}, \dots$ ★ ⑧ $1, 6, 36, \dots$

Write the first three terms of each sequence written below. Assume that the first term is produced when $n=1$.

⑨ $\left\{ \frac{n^2}{2} \right\}$

⑩ $\{2n+4\}$

⑪ $\{ \sqrt[3]{n^2} + 1 \}$

⑫ $\{ \tan(n^4+3) \}$

Determine if each sequence is arithmetic or geometric. Then find an expression for the n^{th} term.

⑬ 8, 18, 28, ...

⑭ -8, 4, -2, ...

⑮ 40, 15, -10, ...

⑯ -4, -24, -144

The following sequences are neither arithmetic nor geometric. Use inductive reasoning to find an expression for the n^{th} term. Assume that the first term is produced when $n=1$.

(17) $\pi-5, 2\pi-5, 3\pi-5, 4\pi-5, \dots$ (18) $\frac{1}{e+2}, \frac{1}{e+4}, \frac{1}{e+6}, \frac{1}{e+8}, \dots$

(19) $\frac{\tan 4}{3}, \frac{\tan 5}{4}, \frac{\tan 6}{5}, \frac{\tan 7}{6}, \dots$ *Hint: n^3 is part of the expression. (20) 11, 18, 37, 74, ...

Write an equation of the n^{th} term of each sequence in recursive form. Also, show the value of the first term.

(21) 8, 18, 28, ... (22) -8, 4, -2, ...

(23) 40, 15, -10, ... (24) -4, -24, -144

Answers

(1) 50

(2) 118.5

(3) $-\frac{2}{3}$

(4) $\frac{9}{4}$

(5) $-\frac{9}{4}$

(6) $\frac{20}{3}$

(7) $\frac{75}{7}$

(8) Because $r=6$ is not between -1 and 1, the sum does not exist.

(9) $\frac{1}{2}, 2, \frac{9}{2}$

(10) 6, 8, 10

(11) $2, \sqrt[3]{4}+1, \sqrt[3]{9}+1$

(12) $\tan 4, \tan 19, \tan 84$

(13) Arithmetic $\{10n-2\}$

(14) Geometric $\{-8(-\frac{1}{2})^{n-1}\}$

(15) Arithmetic $\{-25n+65\}$

(16) Geometric $\{-4(6)^{n-1}\}$

(17) $\{n\pi-5\}$

(18) $\{\frac{1}{e+2n}\}$

(19) $\{\frac{\tan(n+3)}{n+2}\}$

(20) $\{n^3+10\}$

(21) $\{a_1=8, a_n=a_{n-1}+10\}$

(22) $\{a_1=-8, a_n=-\frac{1}{2}a_{n-1}\}$

(23) $\{a_1=40, a_n=a_{n-1}-25\}$

(24) $\{a_1=-4, a_n=6a_{n-1}\}$